

HIGH-LEVEL AGING ESTIMATION FOR FPGA-MAPPED DESIGNS

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ABSTRACT

As the state-of-the-art FPGA devices use the latest advancements in CMOS technology, they also face the reliability challenges of nano-scale CMOS. Transistor aging, mainly due to Bias Temperature Instability (BTI) and Hot Carrier Injection (HCI), is a major reliability issue. In this paper, we present a tool to predict the amount of aging-induced degradation for designs mapped to FPGA devices, to help the designers, in an early phase of the design flow, to choose the appropriate mapping and/or optimization efforts, in order to prolong the lifetime of their FPGA-mapped designs. The tool is based on system level abstractions for both BTI and HCI device level models, in addition to implicit device-level information existed in the power and timing reports provided from the FPGA's vendor tools. A case study of using the tool to explore different designs and mapping options shows that aging of the FPGA device is dependent on the design loaded onto it. Furthermore, different mappings and optimizations of the same circuit can result in different aging rates.

1. INTRODUCTION

The property of FPGA devices of having a regular and highly scalable structure, together with the increasing demands on high performance, low power and reconfigurable circuits, pushes the FPGA manufacturers to exploit the latest advancements in scaled CMOS technology. Nowadays, the state-of-the-art FPGA series from the major vendors use the latest 28 nm high- κ /metal gate process technology [1, 2]. Recently, FPGAs which use a 22 nm process technology and operate with frequencies up to 1.5 GHz have been announced [3]. This puts the FPGA devices in the same line together with the ASIC circuits, facing the reliability challenges accompanied with the technology downscaling at nano-scale.

Transistor aging due to *Hot Carrier Injection* (HCI), and *Bias Temperature Instability* (BTI), is considered one of the major reliability issues at nano-scale [4, 5, 6]. BTI has two correspondents: the negative one (NBTI) that affects PMOS transistors, and the positive one (PBTI) that affects NMOS

transistors. In the past, the effect of PBTI was negligible in comparison to NBTI. However, since the introduction of the high- κ /metal gate materials its effect becomes comparable [6]. The main effect of aging represents in an increase in transistor's threshold voltage and a decrease in the carrier mobility, which leads to a slowdown in transistor's switching speed. These can lead to higher (timing) failures over lifetime and ultimately a reduction in operational lifetime. As FPGA devices face this challenge as well, the designers for FPGA-based circuits have to take care of the amount of expected aging-induced degradation, within the useful lifetime of the device, to achieve reliable circuit operation.

To make the FPGA-based circuit designers capable of analyzing the impact of transistor aging on their designs, they have to consider BTI and HCI-based aging models. In this case, the aging of the mapped design depends both on the FPGA device characteristics and the mapped design netlist. However, the currently available models are at device level [6, 7, 8], and this requires the designer to know the device level information of their targeted FPGA. Aside from the simulation overhead at device level and its complexity, the device level information of the FPGA chip is not available for the FPGA users. This makes the aging estimation, for the designers, an unknown and inaccurate process, and hence, the designers may overestimate it, which results in performance loss, or underestimate it, which may end with an unreliable design. Based on that, the need for fast and accurate aging estimation at system level emerges, even if it is inherently less accurate than the device-level correspondent.

In this paper, we present an aging estimation/prediction tool for designs mapped into state-of-the-art FPGAs. The tool is based on a system level estimation for both BTI and HCI effects. This tool considers both the mapped design (logic-level netlist as well as placement and routing) and FPGA device characteristics. The key point of this tool is that it infers the needed FPGA device information from the available FPGA timing and power reports provided by the FPGA design toolset, and also from the post-place-and-route simulation model, without the need for detailed device-level/circuit-level manufacturing info/parameters, that is usu-

ally needed for the BTI and HCI models. Moreover, the tool uses both thermal modeling and logic simulation of the FPGA mapped design, to obtain accurate stress/relaxation information, which is needed for aging analysis. The tool is implemented based on the information provided by Xilinx's FPGA design tools. Nevertheless, it can be easily adjusted for other FPGA design tools from other vendors.

The tool can be used to evaluate, predict or estimate the amount of aging-induced degradation, and hence the lifetime of the FPGA-mapped designs. It also identifies aging-vulnerable nodes in the design. Designers can obtain the impact of different designs, mapping styles, usage, etc. on the wear-out (lifetime) of their circuits mapped into the FPGA. So, along with existing toolset for power, performance and area analysis, the designers would be able to co-optimize the wear-out along with other design constraints.

As a case study of the tool usage, we have examined the tool with several ITC'99 testbench circuits using a Virtex-6 FPGA, to explore the effect of the design parameters on the amount of expected circuit aging. The results show that the amount of aging-induced degradation is strongly dependent on the circuit mapped to the FPGA device, so that different circuits loaded to a single FPGA device have different lifetimes. Also, different mapping and optimization algorithms for the same circuit show tendencies of different aging rates.

The rest of this paper is organized as follows: In Section 2 the related works are reviewed, then in Section 3 preliminaries on transistor aging are presented. Afterwards, in Section 4, the proposed aging modeling tool is described. The validation of the abstracted device model followed by the experimental results are discussed in Section 5. At the end, Section 6 concludes this paper.

2. RELATED WORK

There are many papers which discuss the effect of BTI and HCI-induced degradations and their estimations. However, only few works presented a complete framework/tool for degradation assessment. For NBTI degradation assessment in microprocessors, the authors in [9] have presented a framework called *New-age* that takes as input parameters the netlist, the technology node used, the operating conditions of the circuit (voltage, temperature, etc.) and the probabilities of the inputs, to calculate the aging estimation due to NBTI. The estimation is done at gate level. A similar framework is presented in [5]. The authors in [10] have also presented a framework for estimating the Mean-Time-To-Failure (MTTF) due to the effects of NBTI, Time-Dependent-Dielectric-Breakdown (TDDB) and Electro-Migration (EM) in FPGAs. The framework depends on a 65 nm predictive technology model (PTM) to get the needed device level information for the estimation, because the FPGA device level information is proprietary. It was not clear how the circuits that represent the different elements of the FPGA are obtained to perform the

estimation at their transistors.

The main difference of the tool proposed in this paper to the aforementioned frameworks, is that our tool does the aging estimation for both BTI and HCI phenomena at system level yet device/technology level relevant, not at transistor or gate-levels. This eliminates any need for detailed device-level information of the FPGA, as this information is usually not available (proprietary). Furthermore, the estimation at system-level is easier, faster and meets the accuracy needs of the FPGA designers to decide whether a certain mapping / routing / placement technique is better than another in terms of aging.

3. PRELIMINARIES

At nano-meter regime, BTI is considered as one of the most important aging issues. It causes the magnitude of the threshold voltage of transistors to increase, which in turn increases the transistor's switching delay [11]. As a result, the total delay of the functional paths may exceed the timing requirements, which causes the circuit to fail. The model of the NBTI effect on PMOS transistor's switching delay (d) can be estimated and simplified as follows [7, 12]:

$$\Delta d = A_{BTI} \times Y^n \times t^n \times e^{\left(\frac{-E_a}{kT}\right)} \times d_0 \quad (1)$$

where d_0 is the "pre-aging" switching delay, Δd is the increase in the switching delay due to NBTI, A_{BTI} is a technology dependent factor, t is the time (transistor's age), Y is the duty cycle, which represents the ratio of stress time (transistor is ON) to total time, and T is the temperature in Kelvin. n can be 1/6 or 1/4 depending on the fabrication process. k is Boltzmann's constant, and E_a is a fitting parameter. As the long-term PBTI effect is dual to the NBTI one, the same model in (1) is used for estimating the PBTI effect on NMOS transistors [13, 14].

HCI, on the other side, affects mainly the NMOS transistors. It is caused by carrier injection into the gate-dielectric, which results in a change of the current-voltage characteristics [15]. The effect on NMOS transistor's switching delay (d) is empirically found and can be simplified as follows [5, 8, 16]:

$$\Delta d = A_{HCI} \times \alpha \times f \times t^{0.5} \times e^{\left(\frac{-E_b}{kT}\right)} \times d_0 \quad (2)$$

where d_0 is the "pre-aging" delay, A_{HCI} is a technology dependent factor, t is the time, E_b is a fitting parameter, α is the activity factor of the transistor, and f is the frequency. The factor $\alpha \times f$ is known as the activity rate, and is referred to by AR in the rest of this paper.

4. METHODOLOGY

The methodology for implementing the aging estimation tool for FPGAs can be divided into three parts: i) the abstraction

of the device level models of both BTI and HCI, ii) the inference of the required information from the power and timing reports of a certain design as well as the thermal model, and iii) the aging estimation process.

4.1. Aging model abstraction

In order to achieve an accurate estimation for the amount of aging-induced degradation, without the need for detailed device level information, a high level abstraction for the device level models for both BTI and HCI is needed. However, this abstraction must use the most detailed information that can be gained from various sources, such as logic-level simulations, detailed physical design (placement and routing) information, thermal profile and the FPGA's power and timing reports, to achieve the highest possible accuracy. By analyzing the FPGA reports, it can be seen that the finest details are given at signal and logic level. Therefore, the abstraction is done at this level, which is called *node level*. A single node can represent an LUT, a flip-flop, a routing signal, a half adder, etc. (see Fig. 1). This node separation is taken at the granularity of what the timing report can provide. A path in the circuit from an input flip-flop to an output flip-flop consists of a set of nodes. It should be noted that several paths can pass through a single node at the same time (e.g. a 6-inputs LUT can have up to six different paths crossing it).

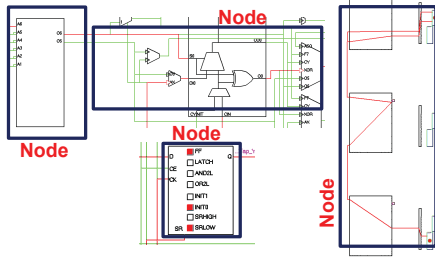


Fig. 1. Different nodes in the FPGA

For a certain node, all the needed information for aging estimation (e.g. signal probabilities, signal activities, initial delays, etc.) can be gained from the FPGA reports for each and every input and output of that node separately. In other words, the device-level details, which are needed for the aging estimation, are already given for the inputs and the outputs of any certain node in the FPGA. However, the device-level details of the node's internals are not known for the FPGA designer. In our proposed model, we get use of this information at the inputs and the outputs of each node to estimate the internal details of that node. The idea is to represent each internal path from an input of a certain node to an output of that node by a series of blocks. Each block is represented by an inverter in the corresponding technology node used in the targeted FPGA, so that it contains the impact of the technology node as well as the aging effect on

both NMOS and PMOS transistors. The number of blocks (m) in the series is determined according to the delay of a representative inverter in the technology node, such that $m = \text{delay of the internal path} / \text{delay of a single inverter}$. In that way, the logic depth of the internal path is represented (see Fig. 2). The validation of this model is discussed in Section 5.1.

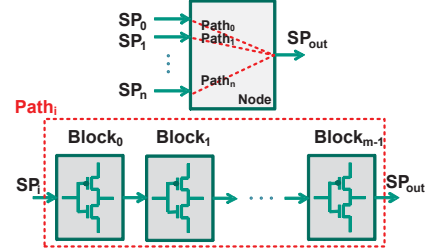


Fig. 2. Different paths through a node and a representation of a single path by a series of blocks

The internal details, at the input of each block, are estimated as a function of the series' input and output (which are given from the FPGA reports for the corresponding node) in addition to the block position in the series. For example, for estimating the NBTI effect, the signal probability (SP), at the gate of a certain transistor, is required to calculate the duty cycle (Y) of that transistor as given in (1). By logic simulation, all signal probabilities, at the inputs/outputs of each node in the circuit, can be calculated. The internal SP for each representative block in Fig. 2, according to the proposed model, can be written as:

$$SP_j = \begin{cases} SP_i + j * \frac{|SP_{out} - SP_i|}{m-1}, & SP_{out} \geq SP_i \\ SP_i - j * \frac{|SP_{out} - SP_i|}{m-1}, & \text{else} \end{cases} \quad (3)$$

where $j \in \{0, 1, 2, \dots, m-1\}$ is the position of the block in the series. Similarly, for HCI, the activity rate (AR) can be read from the power report for each input/output of every node in the design. The internal AR can be then estimated in a similar way like (3). By representing the logic depth, the transistor types, and the delays in the same technology node used in the targeted FPGA, in addition to estimating the internal details (SP s and AR s) from the known information at inputs and outputs of the node, this cascaded block model can better reflect the aging of internal elements of the node.

4.2. Information gathering

The aging estimation tool is based on the abstracted models of BTI and HCI presented in the previous section, and uses these models to calculate the amount of aging-induced degradation. The input parameters for these models are taken from the information provided by:

- A) The timing report to obtain the pre-aging critical paths with their breakdowns of their nodes.
- B) The detailed physical design information, for obtaining the nodes' information and the breakdown of the logic-level netlist to nodes with FPGA coordinates.
- C) Logic simulations for the placed and routed design (with nodes' information) to obtain the duty cycle and activity rates of each input/output of every node.
- D) The thermal model, that uses the power report, physical design information (B) and signal activities (C) to obtain nodes' temperatures.

The tool overview is depicted in Fig. 3. Each part of the tool is discussed in detail in the following subsections.

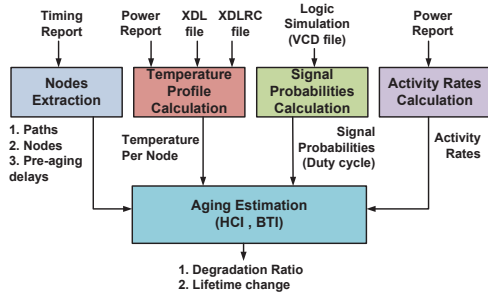


Fig. 3. Overview of the proposed aging estimation tool

4.2.1. Determination of nodes and their delays

The first step of the tool, is to determine the nodes and the pre-aging delay of each path from an input to an output of every node in the finally placed and routed design. This information is taken from the timing report (see Fig. 4). The timing report is generated, in our case, using Xilinx's timing analyzer tool, by specifying a certain number of paths per constraint ($P_{Constr.}$). The value of $P_{Constr.}$ depends on the size of the design, and has to be given such that all the paths that have a slack of $x\%$ are reported (x is the maximum expected degradation during the lifetime of the FPGA, e.g. 15%). A parser is used afterward to extract these paths with their nodes breakdown and also their pre-aging delays.

4.2.2. Determination of activity rates

The activity rate for each input/output of every node (AR) is determined from the power report of the finally placed and routed design. For accurate activity rate calculation, Xilinx's Xpower analyzer needs the *Value Change Dump* file (*.vcd) of the design's simulation. With a logic simulator (e.g. Modelsim), the vcd file can be generated using the post-place-and-route simulation model of the design. However, a testbench that reflects a realistic operation of the desired circuit must be used for the simulation, thus real values

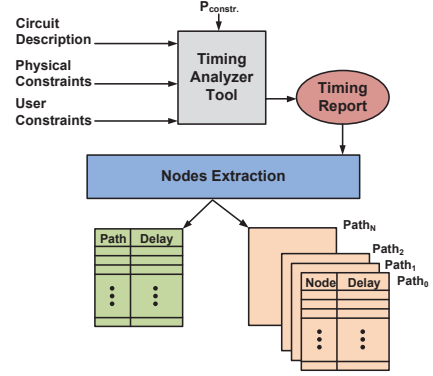


Fig. 4. Determining the nodes in the design and their pre-aging delays

for the activity rates can be reported, which results in a more accurate power estimation of the mapped circuit. A parser is used afterward to read the generated power report, and to extract the nodes and the activity rates at their inputs and outputs. An overview of this process is shown in Fig. 5.

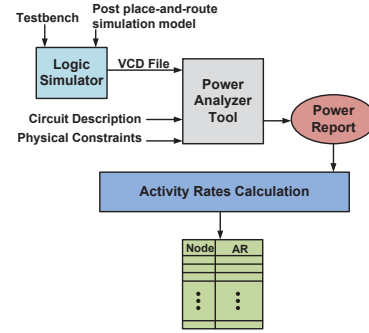


Fig. 5. Calculating the activity rate of the nodes

4.2.3. Determination of signal probabilities

A separate routine is used for this purpose, it parses the vcd file generated in the previous section, then it creates a table with all nodes, and assigns for each node 3 counters to count the times in which the node's input was one, zero or unknown respectively during the simulation. Afterward, the probability of having one at the input is calculated for all nodes, and saved in a separate table. The calculated signal probabilities (SPs) are used afterward to calculate the duty cycle (Y) of each input/output of every node, such that for NBTI: $Y = 1 - SP$, and for PBTI: $Y = SP$.

4.2.4. Temperature profile calculation

A method similar to the one presented in [17] is followed to calculate the temperature profile across the FPGA chip. Fig. 7 shows an overview of this step. We use HotSpot 5.02 [18] for thermal modeling instead of the obsolete HS3d tool used in [17]. The FPGA's floorplan, needed for HotSpot,

is generated using the physical tile locations from the chip resource file (*.xdlrc) and giving the die dimensions. The die in the floorplan is divided into a two-dimensional set of blocks, each block has a dimension of $n \times m$ of physical tiles (we have found that a block size of 5×5 gives the best temperature accuracy). The power report, generated in the step described in Section 4.2.2, is parsed again to extract the power per node values, then the physical locations for each node is obtained from physical mapping info (the design's XDL file), and mapped to the chip resource file (XDLRC file) to find the exact tile place. If a node occupies more than one tile (which is the case for the routing signals), its power is distributed to the tiles, in a way that the tiles that drive a larger number of other signals get more percentage of the power. Afterward, for each block, the values of the power of its tiles are summed up.

For the leakage power, the amount reported in the power report is given for the whole chip and not per node, as the case of the dynamic power. In [17], there was no clue how the leakage power is distributed across the chip. For that reason, as a first rough estimation, it is chosen to divide the leakage power equally to all the blocks. However, when the temperature results from the simulation are compared with real measurements from an infra-red camera, the results did not match. Therefore, it is necessary to use a leakage-temperature loop model to distribute the leakage power to the blocks, in such a way that the block that has higher temperature gets more percentage of the leakage power than the other blocks. We have used the model in (4) for this purpose [19].

$$T = T_a + \theta \times P \quad (4)$$

where T is temperature, T_a is the ambient temperature, θ is the thermal resistance of the package and P is the total power (leakage + dynamic). At the beginning, the leakage power is distributed equally to the blocks, and then HotSpot is called to get the first per-block temperature estimation. Afterwards, the model in Equation (4) is used to calculate the new per-block leakage values. The percentage of change in the leakage is calculated between the old and the new leakage values, and the original amount of leakage reported in the power report is then redistributed to the blocks according to these percentages. In that way, the amount of the reported leakage power does not increase. This loop continues until the values converge. A pseudo-code of this process is shown in Fig. 6.

The calculated temperature values are then mapped, in a similar way, back to calculate the temperature of each node ($T_{(node)}$).

4.3. Aging estimation

At this stage, all the necessary pieces of information for applying the abstracted BTI and HCI models to the nodes are

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1. Read  $P_{leakage}^{total}$ ,  $n \leftarrow No.of.blocks$ 
2.  $P_{leakage}^{block} \leftarrow P_{leakage}^{total} / n$ 
3. Do
4.   Call HotSpot
5.   Read Per_block Temperatures ( $T_{block}$ )
6.   For each  $block_i$ 
7.      $(P_{leakage}^{block_i})_{new} \leftarrow \frac{T_{block_i} - T_a}{\theta} - P_{dynamic}^{block_i}$ 
8.      $factor_i \leftarrow (P_{leakage}^{block_i})_{new} / (P_{leakage}^{block_i})_{old}$ 
9.   End For
10.   $Factors \leftarrow \sum_{i=0}^n factor_i$ 
11.   $P_{leakage}^{factor} \leftarrow P_{leakage}^{total} / Factors$ 
12.  For each  $block_i$ 
13.     $P_{leakage}^{block_i} \leftarrow factor_i \times P_{leakage}^{factor}$ 
14.     $P_{block_i} \leftarrow P_{leakage}^{block_i} + P_{dynamic}^{block_i}$ 
15.  End For
16. While  $\Delta T_{block} > \delta$ 

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Fig. 6. Leakage-temperature loop calculation

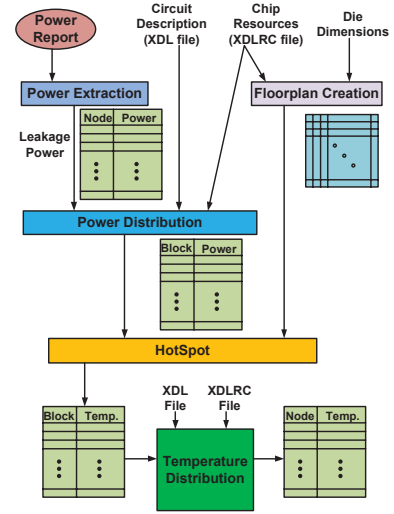


Fig. 7. Temperature calculation using HotSpot

available, and the aging estimation can be made. The estimation process can be simply summarized by three main steps: i) finding the amount of change in node's delay due to aging ($\Delta d_{(node)}$), then ii) calculating, for each node, the post-aging delay ($d_{0(node)} + \Delta d_{(node)}$). Afterward, iii) finding the post-aging critical path delay ($\tilde{d}_{critical}$), and compare it to the pre-aging critical path delay ($d_{critical}$), to determine the percentage of performance loss due to aging. The details are as follows:

- A) Each path in the design P_i contains nodes g_1^i to g_N^i , and each node g_j^i has a pre-aging delay $d_{0(node_j^i)}$. This information is already stored in the table presented in Section 4.2.1.
- B) The pre-aging delay of each path P_i is calculated from $d_0^i = \sum_{j=1}^N d_{0(node_j^i)}$

- C) The pre-aging critical delay $d_{critical}^i$ is the maximum d_0^i for all paths.
- D) Now, for each node g_j^i : i) Find its activity rate $AR_{(node_j^i)}$ from the table in Section 4.2.2. ii) Find its duty cycle $Y_{(node_j^i)}$ from the table in Section 4.2.3. iii) Find its temperature $T_{(node_j^i)}$ from the table in Section 4.2.4.
- E) The amount of change in node's delay due to aging ($\Delta d_{(node)}^i$) can be then calculated for each node. For HCI: i) For the maximum frequency of the targeted FPGA ($AR = max$), assume $x\%$ maximum (worst-case) HCI delay increase in z years¹, under ambient temperature T , and for a representative inverter delay d_0 . Solve in (2) to obtain A_{HCI} . ii) For each node g_j^i , generate the representative series of blocks as discussed in Section 4.1. iii) Having the activity rates at the input and output of the node, calculate for each block the internal activity rate AR similar to (3), then calculate the change in the block delay using (2). iv) The summation of the block delays gives the new delay of the node due to HCI.
- F) Similarly, for BTI: i) For $SP = 1$ (NBTI) or $SP = 0$ (PBTI) Assume $y\%$ maximum (worst-case) BTI delay increase in z years¹, under ambient temperature T , and for a representative inverter delay d_0 . Solve in (1) to obtain A_{BTI} . ii) For each node g_j^i , generate the representative series of blocks as discussed in Section 4.1. iii) Having the signal probabilities at the input and output of the node, calculate for each block the internal signal probability SP as in (3), then calculate the change in the block delay using (1). iv) The summation of the block delays gives the new delay of the node due to BTI.
- G) Afterward, for each node g_j^i , the post aging delay is simply calculated $d_{0(node_j^i)} + \Delta d_{(node_j^i)}$.
- H) At the end, for each path P_i , the post aging delay is calculated $d^i = \sum_{j=1}^N (d_{0(node_j^i)} + \Delta d_{(node_j^i)})$
- I) The post-aging critical delay $\tilde{d}_{critical}$ is the maximum d^i for all paths.

Having both the pre-aging critical delay ($d_{critical}$), and the post-aging critical delay ($\tilde{d}_{critical}$) of the FPGA-mapped design, the percentage of delay degradation due to aging can be simply calculated $aging\ rate = \frac{\tilde{d}_{critical}}{d_{critical}} - 1$.

5. EXPERIMENTAL RESULTS

5.1. Validation of the abstracted aging model

To validate the model proposed in Section 4.1, an HSPICE device level simulation, for a 2-input LUT and for a half

¹The values for x, y and z can be taken from user, or alternatively, from real measurements (e.g. done in [20])

adder circuit, is carried out using a 22 nm predictive technology model. The same circuits and method presented in [21] are used to emulate the effect of aging on the transistors. For each input of the LUT a set of 9 different SPs (0.1, 0.2, ..., 0.9) and their permutations (9*9) are tested for each possible LUT configuration, then the relative delay increase is recorded for each case (see Fig. 8(a)). On the other side, using Matlab, all possible SPs of the LUT's output (SP_{output}) are calculated, then for the path that represent the first input of the LUT to the output, the proposed model is applied. Afterward, for each set of both first-input's SP (SP_0) and SP_{output} , the relative delay increase due to NBTI is recorded and compared to the HSPICE results (see Fig. 8(b)). The comparison results show that, for 14 configurations out of the possible 16, and for all possible permutations of input and output SPs , the trends of NBTI degradation between the model and the HSPICE simulation match. For the remaining 2 configurations, only parts of the permutations of input and output SPs yield similar NBTI degradation trends. In total, around 90% of the results match the trend (see Fig. 8).

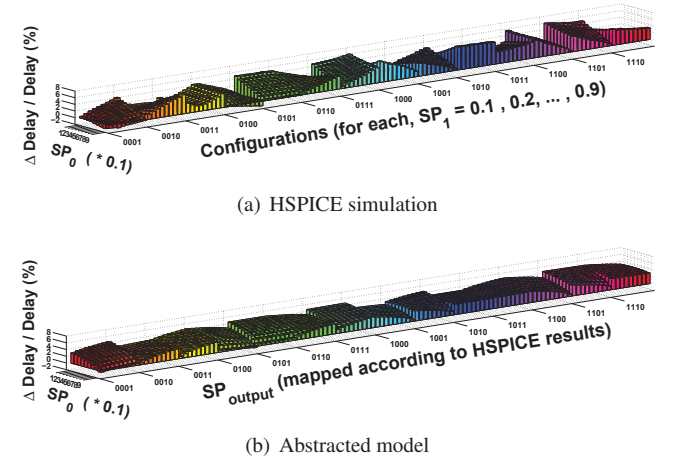


Fig. 8. Validating the abstracted model with HSPICE simulation for pass-transistor-based 2-input LUT, for each input 9 SPs from 0.1 to 0.9 are tested (as in [21]).

It should be noted that the proposed model is to abstract relatively small logic circuits, not large ones. This is thought for the basic elements of the FPGA (nodes). Furthermore, these results show that using the abstracted model alone may not be enough to have an accurate design margin determination. However, for judging whether a certain implementation of the design is better than another, the proposed model gives the needed accuracy. In fact, with the lack of the device level information of the FPGA, this abstraction provides an easy and fast way for aging estimation, which may also be the only available way for FPGA users.

5.2. Case study: influence of mapping and optimization algorithms

The idea is to explore the effect of different design parameters on the amount of estimated aging-induced degradation. Therefore, several ITC'99 testbenches have been tested with different mapping and optimization algorithms. The aging estimation tool is built based on the information provided by Xilinx's FPGA development tools. Virtex6-vlx75t is used as a targeted FPGA for estimating the aging of different designs onto it, because it can fit all the selected testbench circuits. All circuits are forced to be mapped only to LUT and flip-flops, thus a fair comparison can be made. Furthermore, as PBTI effect is similar to NBTI, only NBTI effect is considered in the following. However, the extension of the proposed flow for PBTI is quite straightforward. Three experiments are made for each circuit: the first one is by making the ISE tool choose the best mapping, placing and routing conditions with speed optimization (the default options). The second one is by forcing the placement to a corner of the FPGA with speed optimization. The third experiment is to force area optimization with freely mapping, placing, and routing. In all the experiments, the maximum amount of degradation (worst case), used in Section 4.3 to get the values of A_{BTI} and A_{HCI} , is set to be 15% in 5 years for $SP = 1$ (PMOS transistors always ON $\rightarrow$ worst case NBTI) and $AR = 500$ MHz (maximum frequency $\rightarrow$ worst case HCI). 10000 vectors are tested for each experiment to determine signal probabilities and activities. The degradation ratio due to NBTI and HCI are reported in Table 1 for each experiment. The explanation for the low HCI-induced degradation ratios compared to the NBTI ones is because, for the HCI case, the tested circuits operates with frequencies less than the maximum possible frequency on the FPGA ($AR = 500$ MHz) which is considered the worst case as mentioned before.

It should be noted that the amount of degradation has a direct relation with the circuit behavior (i.e. signal activities and probabilities) as described in Section 4.2.2. Therefore, it is necessary to write a realistic testbench for each circuit to get an accurate circuit behavior. We wrote the testbenches for the ITC'99 circuits, taking into consideration the original functionality of each circuit which are reported in [22]. Nevertheless, for b14, as it represent a sub-circuit of a processor, it was not practical to write a realistic testbench. Therefore, random vector inputs are considered for it.

It is also important to mention that a small change, in the degradation ratio of BTI or HCI-induced aging, can result in a significant change to the operational lifetime of the design. A rough estimation can be made to calculate the percentage of lifetime extension for each case in Table 1. This can be done by considering average properties (temperature, signal probabilities, activity ratios, etc.) of all transistors across the critical path of each design. The percentage of lifetime

extension for each mapping compared to the worst mapping (denoted by "Reference") of every testbench circuit is reported in columns "Lifetime extension" in Table 1 for both NBTI and HCI.

Actually, several conclusions can be drawn from the results in Table 1:

- Aging of the FPGA device (both the NBTI-induced and the HCI-induced), and hence the lifetime, is significantly dependent on the design loaded onto it. This can be clearly seen from the degradation ratios column for both NBTI and HCI.
- Different mappings of the same circuit can results in relatively different aging rates.
- Changing the mapping and/or the optimization algorithm can significantly affect the lifetime of the design.
- It cannot be concluded whether the area-optimized mapping is always better in terms of aging compared to delay-optimized or vice versa (for both NBTI and HCI). It depends on and differs for each application.

6. CONCLUSION

Aggressive device downscaling at nano-scale in CMOS technology is one of the main drives for the continuation of Moore's law. The reliability challenges accompanied with the downscaling are also faced by the state-of-the-art FPGA devices as they use the latest advancements in CMOS technology. Transistor aging, mainly due to Bias Temperature Instability (BTI) and Hot Carrier Injection (HCI), is a major reliability issue. Due to the lack of device level information of FPGA devices, the FPGA users are unable of using the device level models of BTI and HCI to predict the amount of aging-induced degradation for their FPGA-mapped designs.

In this paper, we have presented a tool to predict the amount of aging-induced degradation for designs mapped to FPGA devices, based on system level abstractions for both BTI and HCI device level models. The tool extracts the necessary information from the implicit data existed in the power and timing reports provided from the FPGA's vendor tools. Using the proposed tool, several experiments have been made to explore the effect of different designs and mapping options. The results show that aging of the FPGA device is dependent on the design loaded onto it. Furthermore, different mappings of the same circuit can result in different aging rates.

7. REFERENCES

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Table 1. Results of investigating different mapping options for each design and the effect on both HCI/NBTI-induced aging

Circuit	Experiment	Area (LUT / FF)	Pre-aging critical delay	Post-aging critical delay (NBTI)	Post-aging critical delay (HCI)	Degradation ratio (NBTI)	Lifetime extension ratio	Degradation ratio (HCI)	Lifetime extension ratio
b04	Speed optimized	111 / 67	4.108 ns	4.622 ns	4.233 ns	12.51%	75.60%	3.04%	39.26%
	Forced placement	112 / 67	4.080 ns	4.617 ns	4.226 ns	13.16%	29.60%	3.58%	Reference
	Area optimized	106 / 59	5.341 ns	6.075 ns	5.481 ns	13.74%	Reference	2.62%	143.94%
b05	Speed optimized	133 / 41	2.777 ns	3.056 ns	2.794 ns	10.05%	120.33%	0.61%	54.42%
	Forced placement	133 / 41	2.766 ns	3.083 ns	2.787 ns	11.46%	Reference	0.76%	Reference
	Area optimized	96 / 34	3.169 ns	3.497 ns	3.185 ns	10.35%	84.30%	0.50%	159.06%
b11	Speed optimized	68 / 31	2.142 ns	2.442 ns	2.225 ns	14.01%	Reference	3.87%	98.56%
	Forced placement	68 / 31	2.263 ns	2.569 ns	2.390 ns	13.52%	23.47%	5.61%	Reference
	Area optimized	63 / 19	2.603 ns	2.966 ns	2.711 ns	13.95%	2.61%	4.15%	110.44%
b12	Speed optimized	256 / 123	2.348 ns	2.618 ns	2.365 ns	11.50%	60.15%	0.72%	21.47%
	Forced placement	256 / 123	2.370 ns	2.654 ns	2.389 ns	11.98%	25.05%	0.80%	Reference
	Area optimized	192 / 119	3.023 ns	3.399 ns	3.043 ns	12.43%	Reference	0.66%	87.29%
b14	Speed optimized	856 / 216	6.451 ns	7.256 ns	6.604 ns	12.48%	31.67%	2.37%	Reference
	Forced placement	856 / 216	7.173 ns	8.110 ns	7.285 ns	13.06%	Reference	1.56%	156.38%
	Area optimized	765 / 162	8.256 ns	9.305 ns	8.320 ns	12.70%	18.04%	0.78%	1096.71%
FIR Filter	Speed optimized	1086 / 515	3.513 ns	4.026 ns	3.574 ns	14.60%	2.73%	1.74%	15.24%
	Forced placement	1086 / 515	3.784 ns	4.339 ns	3.843 ns	14.67%	Reference	1.32%	53.85%
	Area optimized	962 / 499	4.311 ns	4.908 ns	4.400 ns	13.85%	41.18%	2.06%	Reference

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